DFA Fixed Income Investment Strategy

Comprehensive Summary of Principles of Dimensional Investing

EXECUTIVE SUMMARY

Dimensional Fund Advisors has managed fixed income since 1983, developing a scientific, transparent, and process-driven investment approach based on rigorous theoretical and empirical research. The core philosophy: current market prices integrate and reflect the aggregate views of market participants, containing reliable information about systematic differences in expected returns among securities.

CORE INVESTMENT PHILOSOPHY

Market Prices Contain Relevant Information

Over 50 years of research demonstrates that market prices provide the best estimate of a security's value. Dimensional systematically uses information in market prices to:

- Identify systematic differences in expected returns
- Monitor and manage risk in real-time
- Enable efficient, cost-effective trading execution

BOND EXPECTED RETURN DECOMPOSITION

The expected return of a bond can be decomposed into three components:

Component	Description	Known/Observable
1. Current Yield	The bond's current price or yield	YES - Observable
2. Expected Capital Appreciation	Expected price change from moving along the yield curve	YES - Observable
3. Unexpected Yield Changes	Future changes in the yield curve itself	NO - Unpredictable

Components 1 and 2 contain reliable information and can be used to assess expected returns. Component 3 adds noise but 40+ years of research shows there is no reliable way to predict changes in interest rates.

TERM PREMIUM: DURATION STRATEGIES

Using the Yield Curve to Enhance Returns

DFA uses information in current market prices to pursue higher expected returns by dynamically varying portfolio duration. This 'variable maturity' approach responds to yield curve shape:

Upward Sloping Yield Curve:

• Longer-term bonds have higher yields

• Expected capital appreciation is positive

• Strategy: Extend duration to upper end of eligible range

Inverted Yield Curve:

• Shorter-term bonds have higher yields

• Expected capital appreciation would be negative for longer bonds

• Strategy: Shorten duration, overweight shorter-term bonds

Term Spreads and Premiums

Historical evidence (1976-2016) shows a reliable relationship between term spreads and future term premiums:

Term Spread	Average Monthly Premium	T-Statistic
All months	6 bps	2.57
Spread > 25 bps	8 bps	2.64
Spread > 50 bps	15 bps	3.33

Source: Bloomberg Barclays Indices (Intermediate minus 1-3 Year), 1976-2016

CREDIT PREMIUM: QUALITY STRATEGIES

Variable Credit Approach

DFA uses information in credit spreads to dynamically vary credit quality allocation. The expected return of a credit bond accounts for default probability:

Expected Return = $(1 - P_{default}) \times (\text{Yield} + \text{Term} + \text{Future Change}) - P_{default} \times (1 - E(\text{Recovery Rate}))$

Credit Spreads and Premiums

Historical evidence (1973-2016) demonstrates a reliable relationship between credit spreads and future credit premiums:

Credit Spread	Average Monthly Premium	T-Statistic
All months	7 bps	2.09
Spread > 50 bps	10 bps	2.96
Spread > 100 bps	15 bps	3.23

Source: Bloomberg Barclays Indices (Credit minus Government), 1973-2016

Key Implication: Investors need to remain invested in credit when spreads are wide, as historically this is when the majority of credit premiums have been realized. Flights to quality during wide spread periods decrease the likelihood of capturing credit premiums.

Real-Time Credit Quality Monitoring

DFA supplements official credit ratings with market-implied ratings. Research shows bonds trading at higher yields than their peer credit group are significantly more likely to experience future downgrades:

Yield Category	After 3 Months	After 6 Months	After 12 Months
All Bonds	2.5%	4.8%	7.2%
Yields Near Lower Rating	5.1%	12.1%	16.6%
Yields Above Lower Rating	10.1%	23.5%	32.0%

Source: Bloomberg Barclays US Aggregate Index, Investment Grade Corporates, 2010-2016

GLOBAL DIVERSIFICATION

Benefits of International Fixed Income

The US represents only about 30% of the global investment grade bond market. Global bonds enhance diversification and expected returns through:

- **Imperfectly correlated term structures** across countries reduce volatility
- **Multiple yield curves** provide more opportunities for term premiums
- **Currency hedging** eliminates currency risk while accessing foreign term premiums

Empirical Evidence: Volatility Reduction

Index	Avg Monthly Return	Monthly Std Dev	Volatility Reduction
US Only (1-3 Year)	0.41%	0.53%	Baseline
Global (Hedged, 1-3 Year)	0.41%	0.41%	22%

Source: Citigroup WGBI USD Indices, 1985-2016. Same return, 22% lower volatility.

Currency Hedging Mechanics

When a foreign bond's currency exposure is hedged, expected return becomes:

Hedged Return ≈ Local Bond Return + Forward Currency Premium

Forward currency premiums contain information about differences in expected returns between hedged and unhedged positions. This allows DFA to use exchange rate information to inform hedging decisions and increase expected returns.

PORTFOLIO IMPLEMENTATION

Dynamic Daily Process

Portfolio managers implement a systematic daily process that balances expected returns against risks and costs:

1. **Monitor Holdings:** Review current positions and calculate available cash
2. **Compute Expected Returns:** Rank all eligible securities by expected return (net of costs)
3. **Identify Opportunities:** Find securities that increase expected returns within duration/credit/currency constraints
4. **Credit Monitoring:** Use multiple inputs (ratings, prices, CDS, stocks) to assess creditworthiness
5. **Meaningful Turnover:** Only trade when expected return improvement exceeds transaction costs
6. **Submit Orders:** Provide traders with multiple candidate buy/sell orders
7. **Review Execution:** Update holdings and prepare for next day

Portfolio Design Examples

Strategy	Duration Range	Credit Quality	Key Features
One-Year Fixed Income	0-1 year	High quality short-term	• Capital preservation • 3% issuer cap • Minimal default risk
Investment Grade Portfolio	Benchmark relative	AAA to BBB	• Core bond strategy • 3% cap (AAA/AA) • 1% cap (A/BBB) • Global diversification

TRADING EXECUTION: PATIENT FLEXIBILITY

The Value of Trading Flexibility

DFA's investment philosophy creates natural trading flexibility. Because bonds with similar expected returns are close substitutes (subject to diversification), portfolio managers can generate multiple candidate orders. This flexibility allows traders to:

- Participate in natural daily market liquidity without demanding immediacy
- Walk away from unfavorable trades (wide bid-ask spreads)
- Execute at more favorable prices than liquidity-seeking competitors
- Maintain fully invested portfolios despite selectivity

Quantified Trading Advantage

Analysis of 2016 TRACE data demonstrates DFA's execution advantage:

Bond Type	DFA Buy Advantage vs. Adjacent Trades	DFA Sell Advantage vs. Adjacent Trades
Agency Bonds	7 bps lower purchase price	5 bps higher sale price
Corporate Bonds	24 bps lower purchase price	20 bps higher sale price

Source: TRACE eligible trades, January-December 2016. Adjacent trades = transactions immediately before and after DFA trades.

DFA maintains 4-5 potential substitute trades for every executed buy, providing substantial flexibility to avoid demanding immediacy from markets.

TRACE Database Utilization

DFA analyzes TRACE data to identify potential dealer inventory, combining historical and intraday transaction data. This automated process runs throughout the day, helping traders:

- Locate bonds that fit portfolio parameters
- Identify dealers with inventory
- Introduce greater competition into bidding/offering process
- Execute more efficiently with better pricing

RISK MANAGEMENT

Diversification Framework

DFA emphasizes appropriate diversification across multiple dimensions:

- **Issuer diversification:** Issuer caps prevent concentration (e.g., 3% for AAA/AA, 1% for A/BBB)
- **Guarantor diversification:** Guarantor caps manage interconnected risk
- **Industry diversification:** Avoid sectoral concentration
- **Country diversification:** Reduce country-specific risk
- **Currency diversification:** Hedged exposure to multiple yield curves

Managing the Diversification-Return Tradeoff

While concentrating in the highest expected return securities could theoretically maximize returns, diversification helps:

- Reduce impact of unpredictable yield changes (component 3)
- Increase reliability of outcomes
- Manage security-specific risks
- Provide trading flexibility

DFA uses current market prices to effectively balance this tradeoff, pursuing higher expected returns while maintaining appropriate diversification.

KEY PRINCIPLES AND TAKEAWAYS

Principle	Implementation
Market Prices Contain Information	Use yields, spreads, and forward rates to identify systematic differences in expected returns
No Interest Rate Prediction	40+ years of research: no reliable way to predict future interest rate changes. Focus on observable components 1 & 2.
Dynamic Duration (Variable Maturity)	Extend duration when curves are steep, shorten when inverted. Respond to term spread information daily.
Dynamic Credit (Variable Credit)	Increase credit exposure when spreads are wide (higher premiums), decrease when narrow. Use market-implied ratings.
Global Diversification	Access multiple yield curves (30 countries+) with currency hedging. 22% volatility reduction vs. US-only.
Meaningful Turnover	Trade only when expected return improvement exceeds all costs. Maintain flexibility across time and securities.
Patient Trading	Avoid demanding immediacy. Use TRACE data and multiple candidate orders. Execute 7-24 bps better than peers.
Comprehensive Credit Monitoring	Use market prices, ratings, CDS, bid-ask spreads, and equity prices. Update credit views in real-time.

CONCLUSION

Dimensional's fixed income approach represents a systematic, research-driven methodology built on five decades of financial economics research. Rather than attempting to outguess markets by predicting future interest rate movements, DFA uses the reliable information contained in current market prices to:

1. **Identify opportunities:** Systematically find bonds with higher expected returns along duration, credit, and currency dimensions
2. **Manage risk:** Monitor credit quality in real-time using market-based information and maintain appropriate diversification
3. **Execute efficiently:** Trade with patience and flexibility, avoiding the high costs of demanding immediacy

This integrated approach—combining research, portfolio design, management, and trading—has enabled DFA to deliver competitive investment solutions for over 35 years. The framework is transparent, systematic, and continuously evolving as markets develop and understanding improves.

Three Fundamental Questions

Question	DFA Answer
Can we predict future interest rates?	No. 40+ years of research shows no reliable prediction method. Focus on components 1 & 2 which are observable.
Can we identify bonds with higher expected returns?	Yes. Term spreads, credit spreads, and forward currency premiums contain reliable information about future premiums.
Can we implement efficiently?	Yes. Patient trading with flexibility consistently achieves 7-24 bps better execution than market participants demanding immediacy.

The result is a disciplined, systematic approach that maintains continuous focus on higher expected returns while managing the inevitable tradeoffs between returns, risks, and costs that characterize fixed income investing.

Summary prepared for DFA Milestone 1 exam preparation. Source: Principles of Dimensional Investing, Fixed Income Strategies section, pages 63-101.